Archimate
Business Layer



Who am I?

Mattia Sabella

PhD Candidate in Information Technology Area at DEIB

Topic: Decentralized AI & Data Quality

Contacts: mattia.sabella@polimi.it / linkedin.com/in/mattia-sabella

Archimate Definition

Archimate is a **standardized modeling language** that aims at:

- **Visualize**
- **Describe**
- **Analyze**

What?

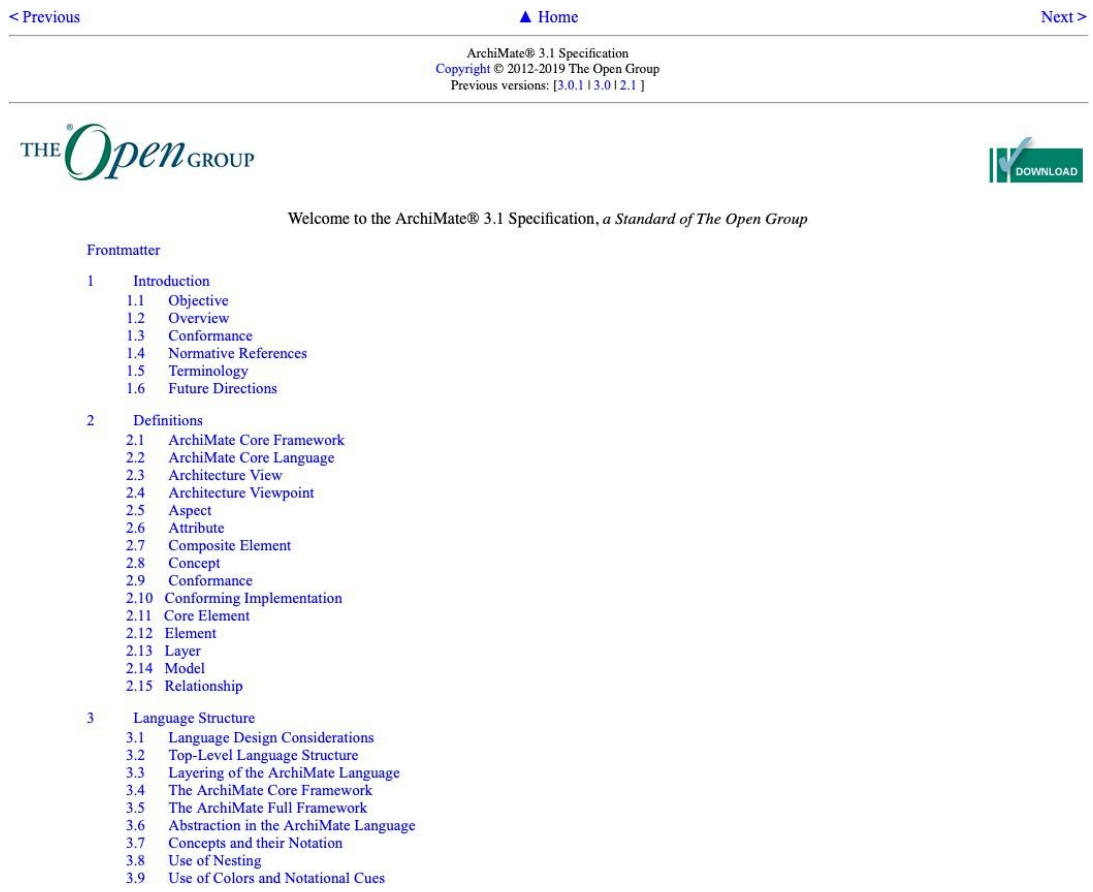




Relationship between
Business processes
Applications
Technology infrastructure

Archimate specification

https://pubs.opengroup.org/architecture/archimate3-doc/



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ArchiMate® 3.1 Specification
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Previous versions: [3.0.1 | 3.0 | 2.1]

THE Open GROUP

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Welcome to the ArchiMate® 3.1 Specification, *a Standard of The Open Group*

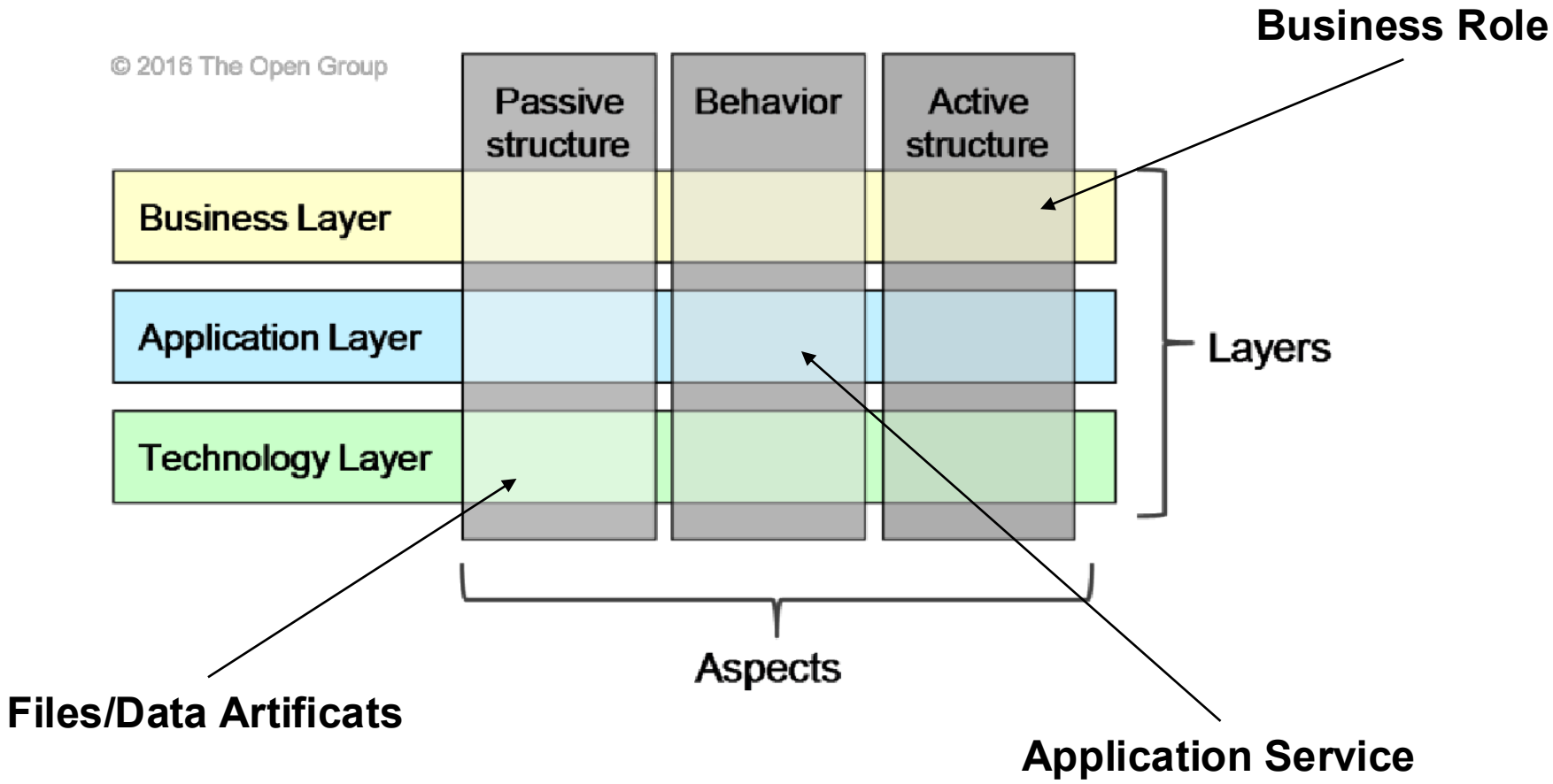
Frontmatter

1 Introduction
- 1.1 Objective
- 1.2 Overview
- 1.3 Conformance
- 1.4 Normative References
- 1.5 Terminology
- 1.6 Future Directions

2 Definitions
- 2.1 ArchiMate Core Framework
- 2.2 ArchiMate Core Language
- 2.3 Architecture View
- 2.4 Architecture Viewpoint
- 2.5 Aspect
- 2.6 Attribute
- 2.7 Composite Element
- 2.8 Concept
- 2.9 Conformance
- 2.10 Conforming Implementation
- 2.11 Core Element
- 2.12 Element
- 2.13 Layer
- 2.14 Model
- 2.15 Relationship

3 Language Structure
- 3.1 Language Design Considerations
- 3.2 Top-Level Language Structure
- 3.3 Layering of the ArchiMate Language
- 3.4 The ArchiMate Core Framework
- 3.5 The ArchiMate Full Framework
- 3.6 Abstraction in the ArchiMate Language
- 3.7 Concepts and their Notation
- 3.8 Use of Nesting
- 3.9 Use of Colors and Notational Cues

Archimate CORE framework



Layers

- **Business layer:**
  offer products and services to external costumers.

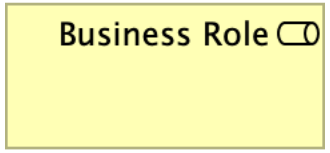
- **Application layer:**
  support the business layer with application services – realized by software applications

- **Technology layer:**
  offers infrastructure services needed to run applications, realized by computer and communication

Active elements



A business entity that is capable of performing behavior



The responsibility for performing specific behavior, to which an action can be assigned

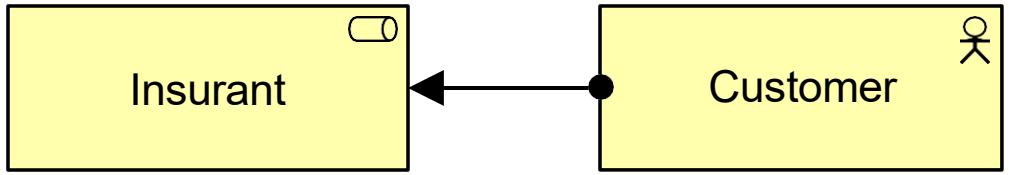


A point of access where a business service is made available to the environment

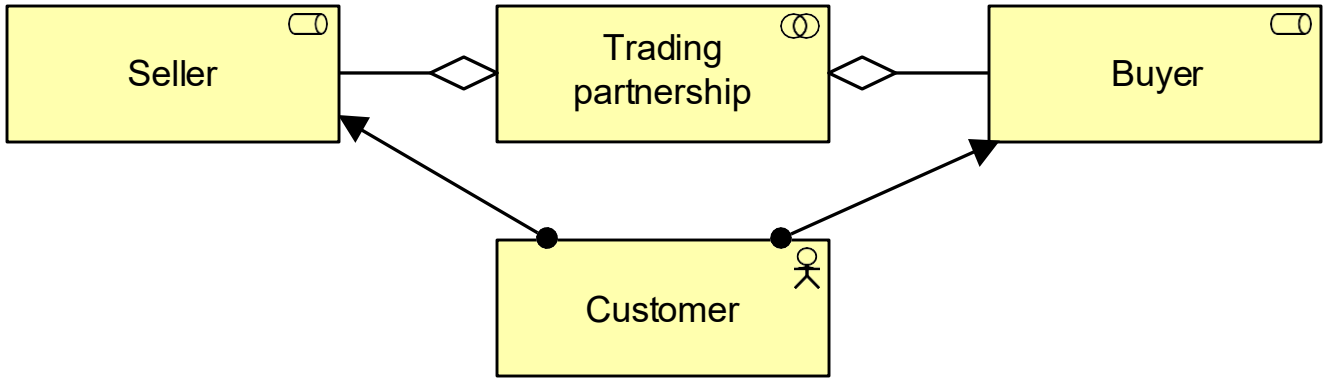


An aggregate of two or more business internal active structure elements that work together to perform collective behavior

Roles and actors



Interaction through collaboration



Interaction through service



Behavioral elements



A sequence of business behaviors that achieves a specific outcome



A collection of business behaviors based on a chosen set of criteria, aligned to an organization

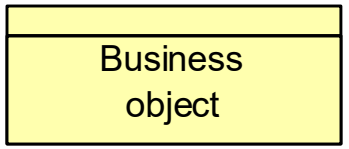


An explicitly defined exposed business behavior

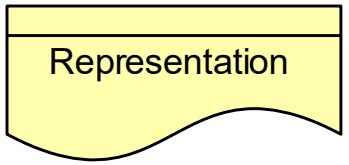


A business behavior element that denotes an organizational state change.

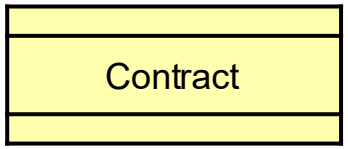
Passive elements



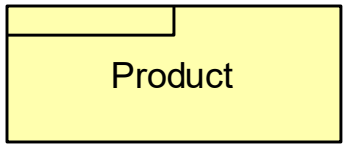
A concept used within a particular business domain



The perceptible form of information carried out by a business object

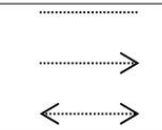
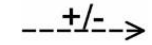


A formal or informal specification of an agreement between a provider and a consumer



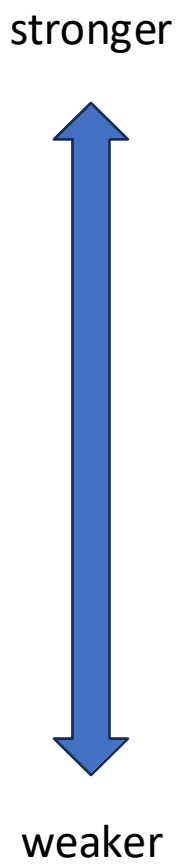
A collection of services and passive structure elements, which is offered as a whole to customers

Relationships

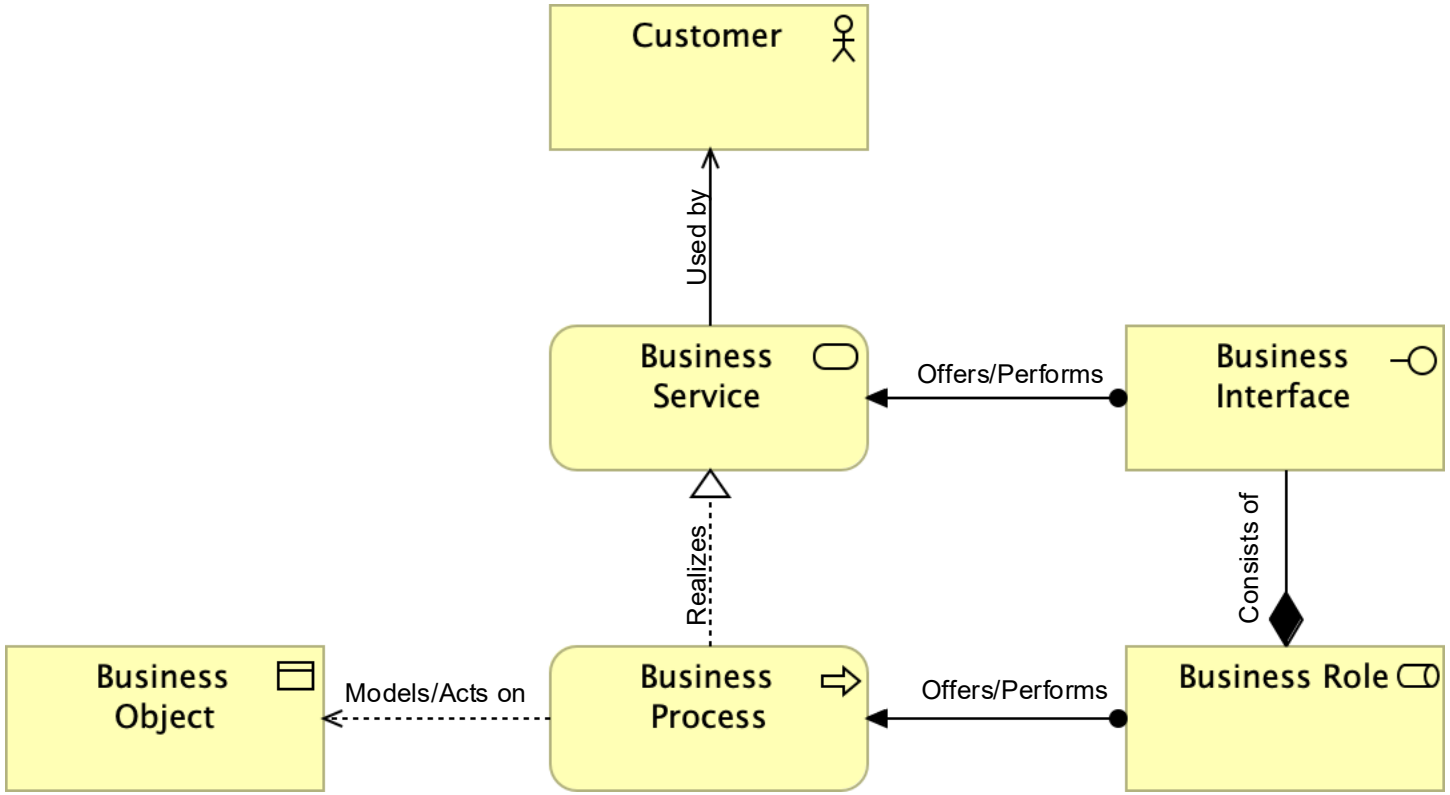
Structural Relationships		**Notation**
Composition	Indicates that an element consists of one or more other concepts.	
Aggregation	indicates that an element groups a number of other concepts.	
Assignment	Expresses the allocation of responsibility, performance of behavior, or execution.	
Realization	Indicates that an entity plays a critical role in the creation, achievement, sustenance, or operation of a more abstract entity.	
Dependency Relationships		**Notation**
Serving	Models that an element provides its functionality to another element. "Used by"	
Access	Models the ability of behavior and active structure elements to observe or act upon passive structure elements.	
Influence	Models that an element affects the implementation or achievement of some motivation element.	
Dynamic Relationships		**Notation**
Triggering	Describes a temporal or causal relationship between elements.	
Flow	Transfer from one element to another.	
Other Relationships		**Notation**
Specialization	Indicates that an element is a particular kind of another element.	
Association	Models an unspecified relationship, or one that is not represented by another ArchiMate relationship.	
Junction	Used to connect relationships of the same type.	

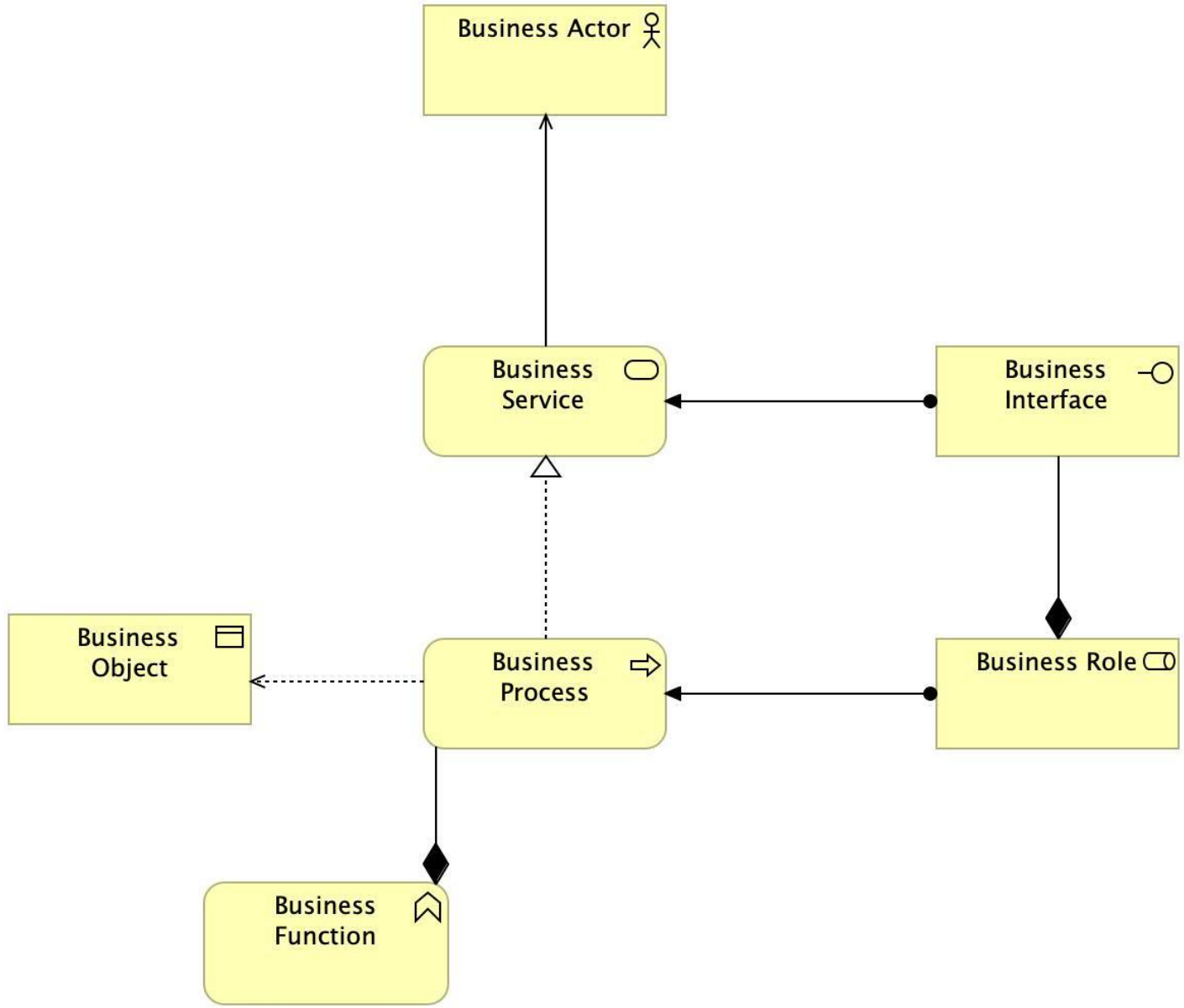
Strengths of relationships

Structural R	**Notation**
Composition	
Aggregation	
Assignment	
Realization	
Dependency	**Notation**
Serving	
Access	
Influence	+/-
Dynamic Rel	**Notation**
Triggering	
Flow	
Other Relati	**Notation**
Specialization	
Association	



Basic business pattern

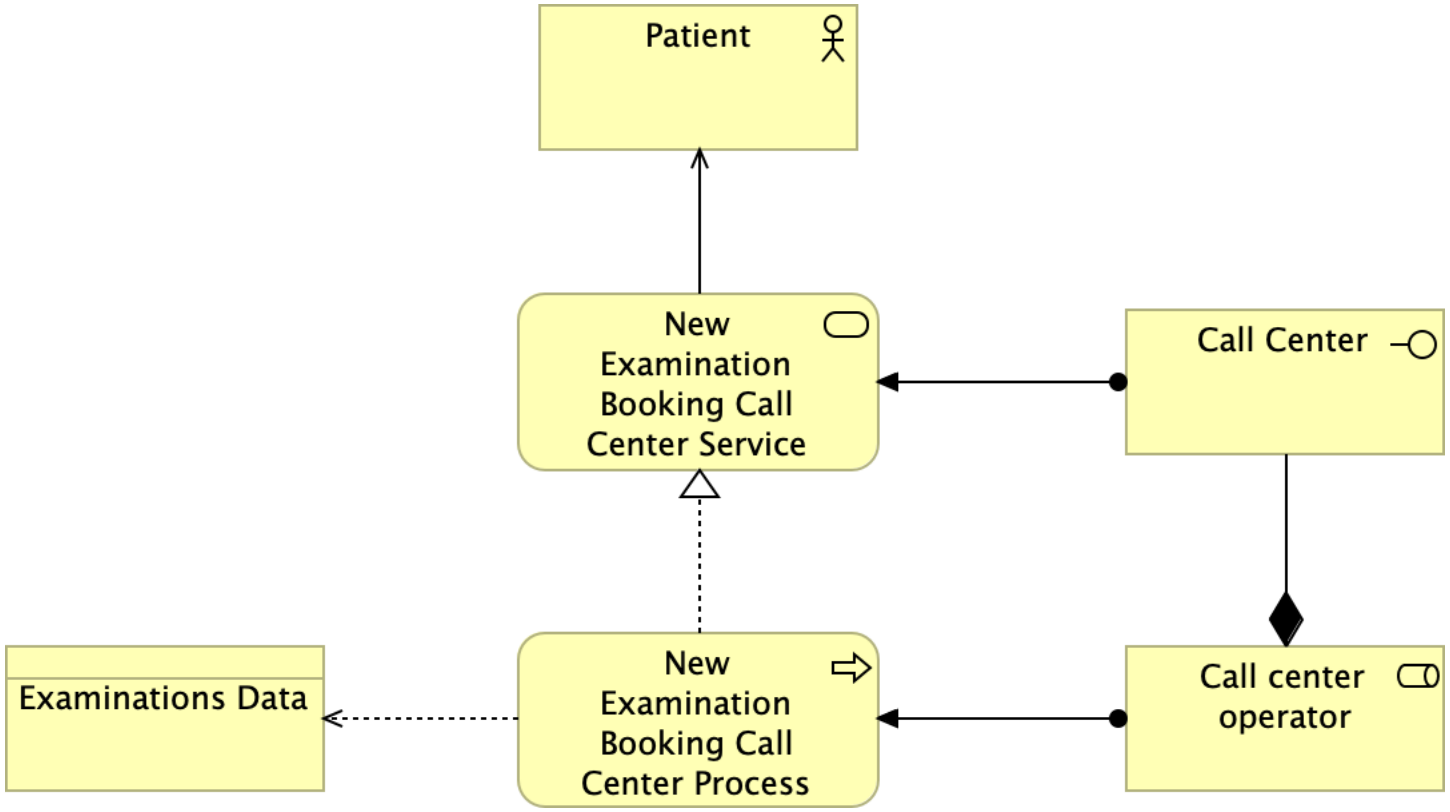



Business Actor
Business Service
Business Interface
Business Object
Business Process
Business Role
Business Function

A first example

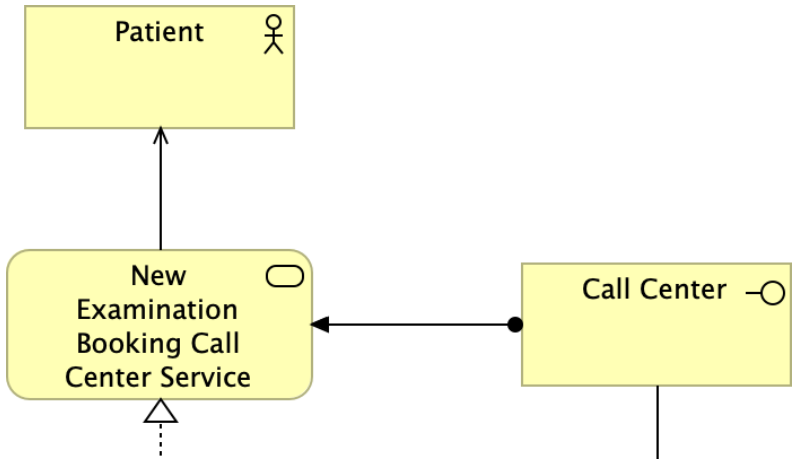
Among the services offered to its patients, the ACME hospital offers a **booking service** for medical visits through a **Call Center**, which allows the **patients** to **book a new medical examination**.

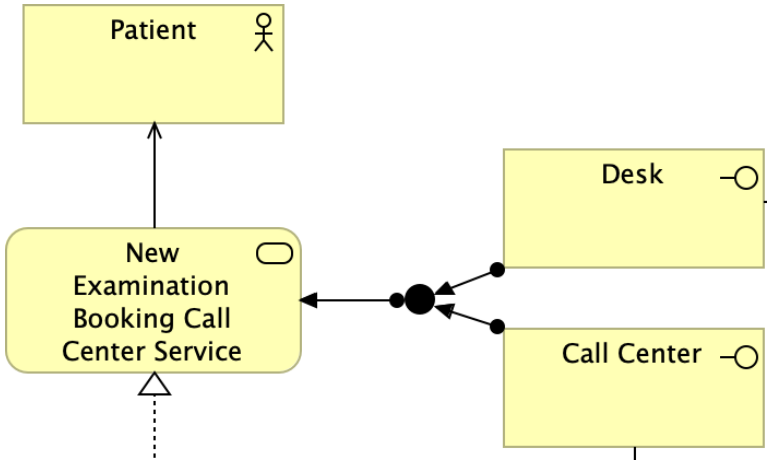
A first example



Above the visible line

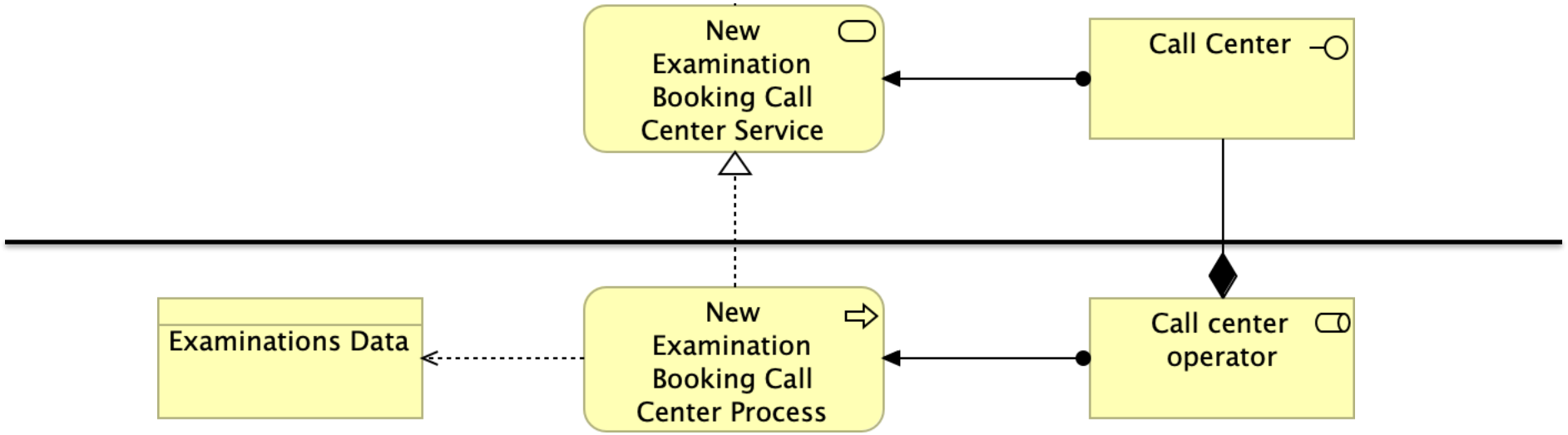
- The user sees a service accessible to a given interface
- Interfaces can be more than one (use junctions)





Behind the visibility line

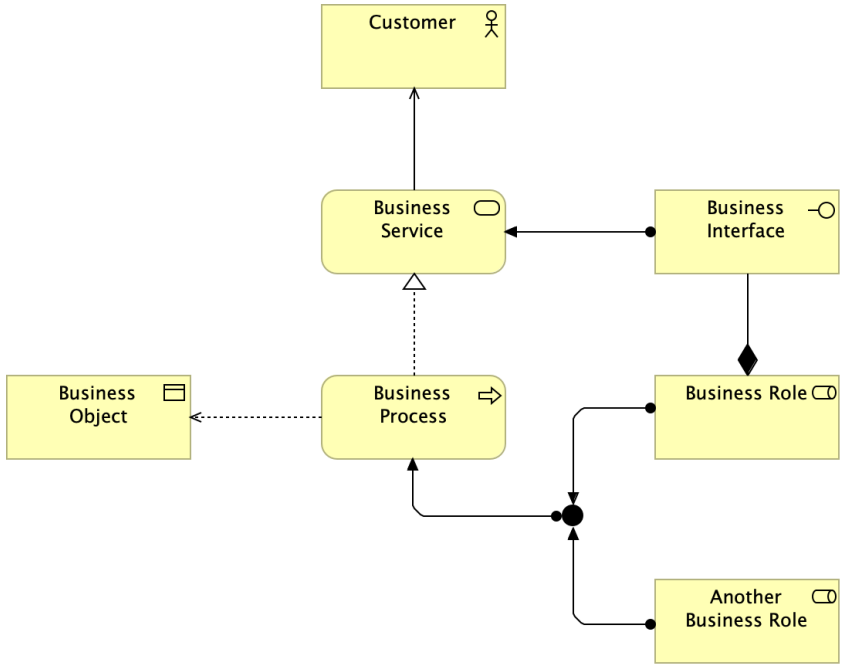
- The process to realize the service is defined along with the actors/roles involved
- The role assigned to the process knows the process

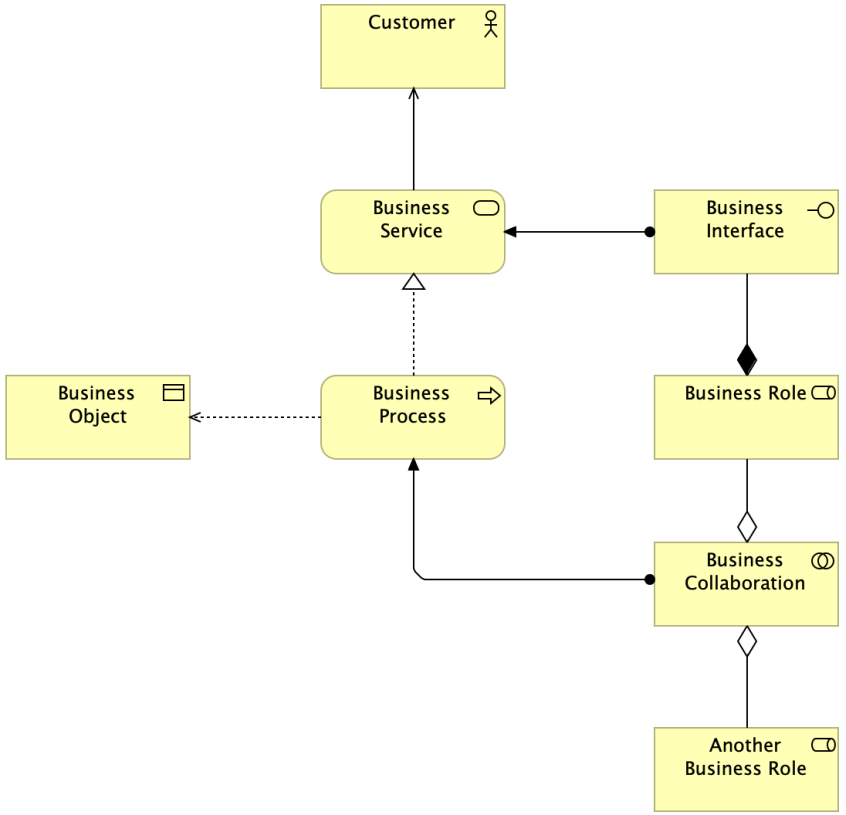


Role assigned to a process

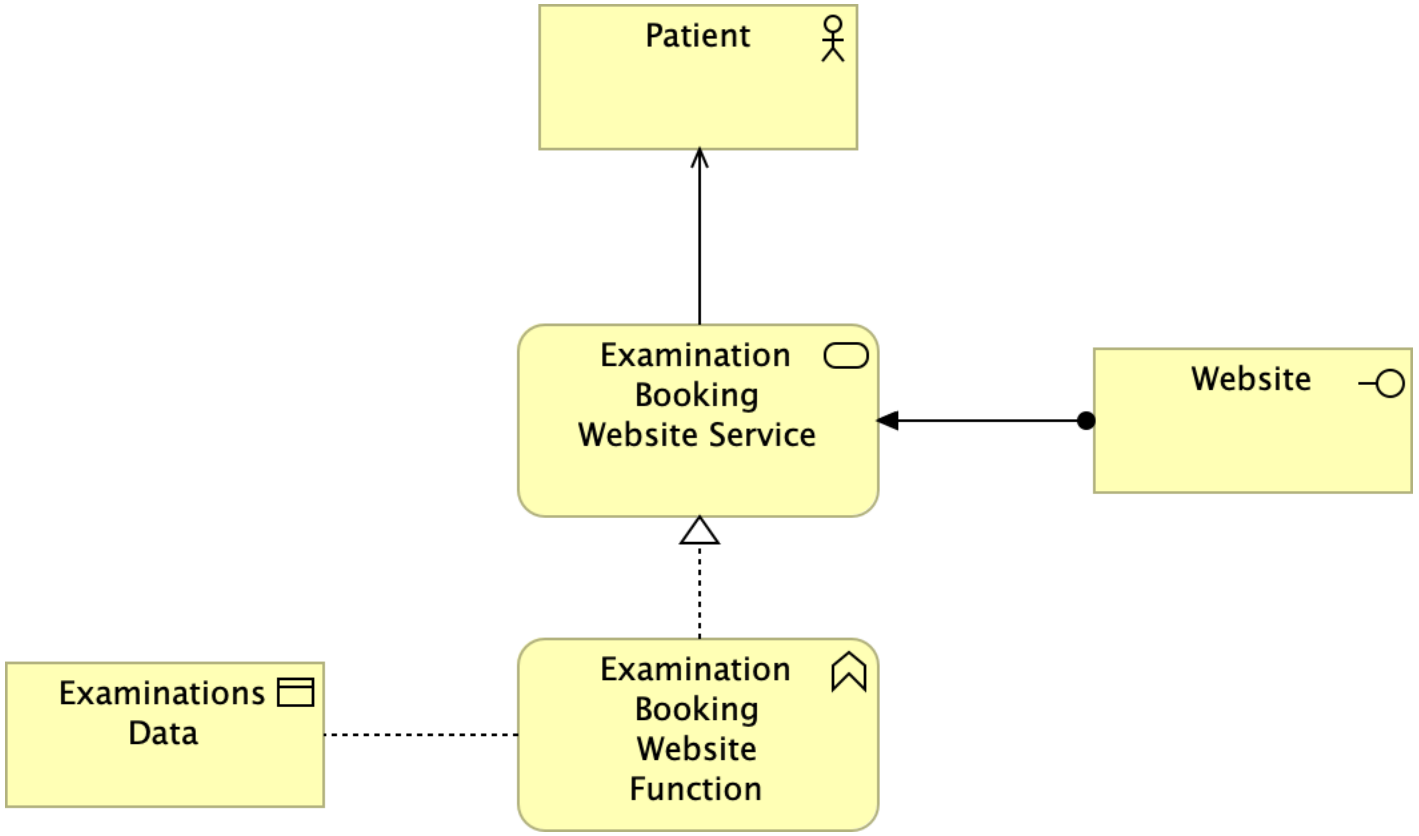
- There must be at most one role assigned to a process
  - Junctions or collaborations can be introduced to manage multiple assignments
  - In case of no assignments, the process is considered as fully automated

More than one responsible





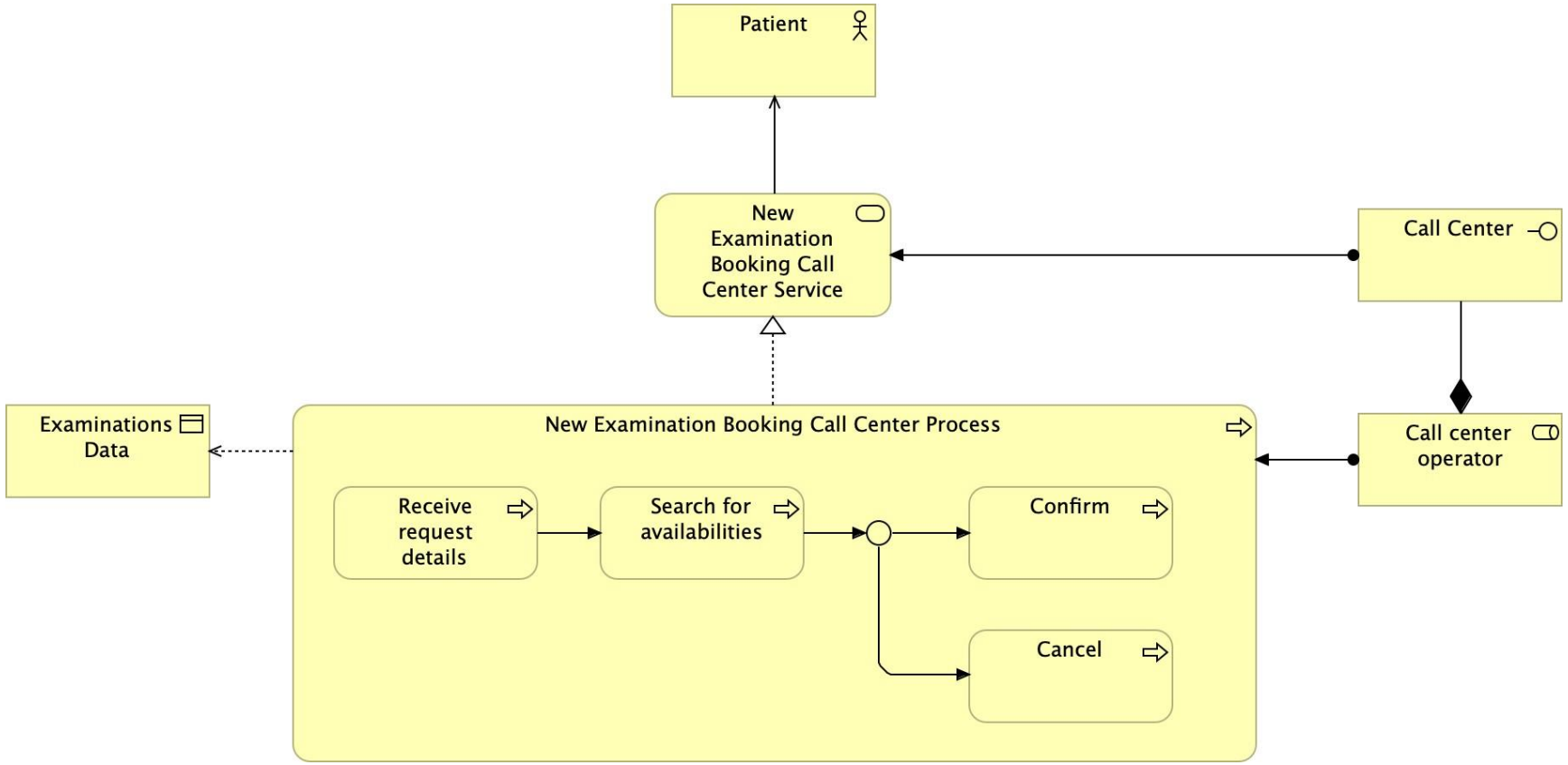
A self-serve service



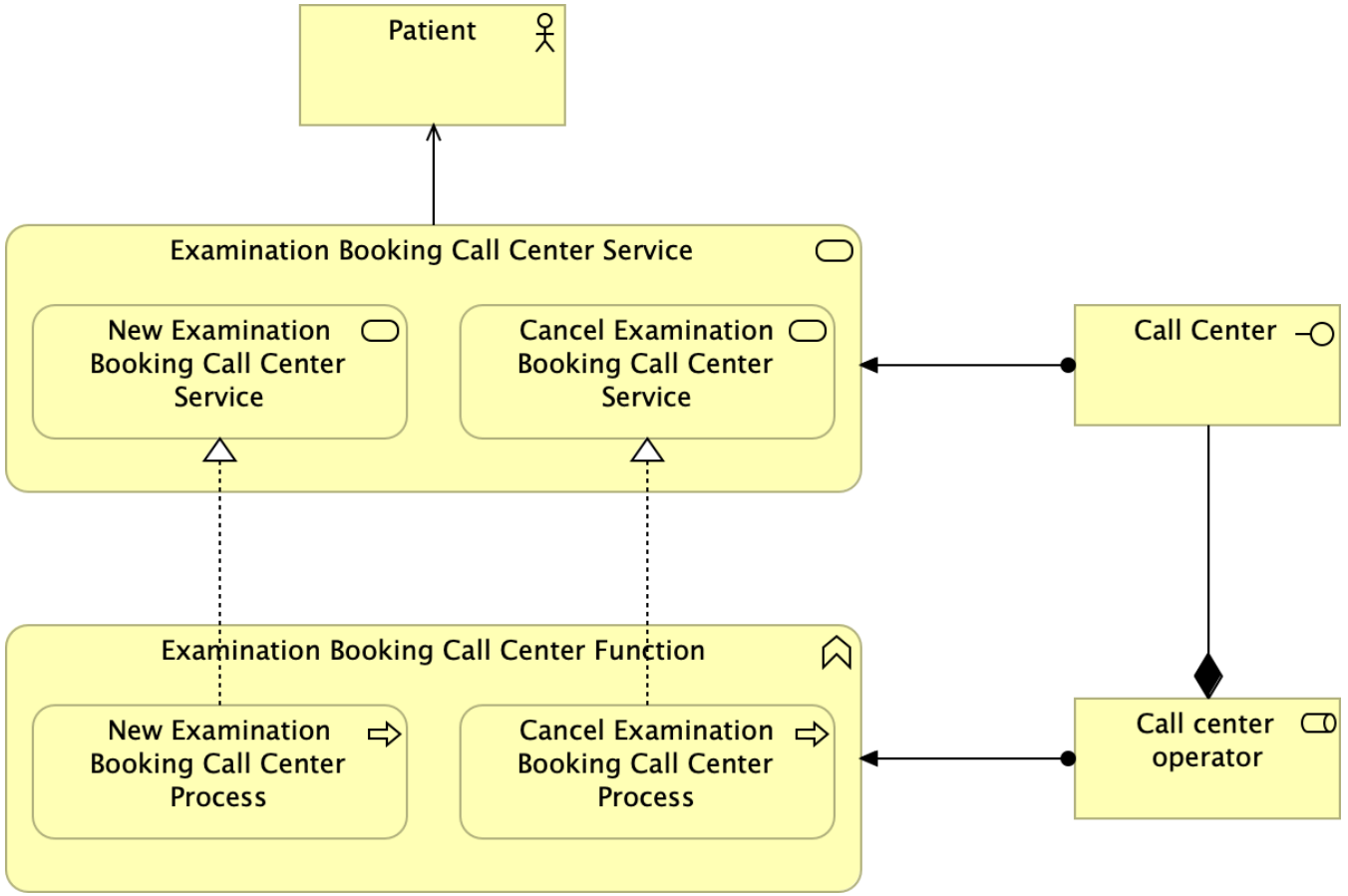
On functions and processes: business layer

- Functions and processes are performed by a single role
  - If more roles are involved use either a junction or an interaction (which is performed by a collaboration) to make it more explicit
- Functions and processes can be composed of other behavioural elements
- Functions are used when:
  - The composing elements have something in common (e.g., the role, the resources)
- Processes are used when:
  - The composing elements collectively achieve the same goal

Process specification



Function specification



Software

Software per Windows:

Stencil for Microsoft Visio (free for Polimi Students):

https://architecture-center.com/blog/archimate-3-0- stencil-set-for-visio.html

Software cross platform:

Archi (recommended):

https://www.archimatetool.com

Web Applications:

Signavio (free for Polimi students):

http://academic.signavio.com/p/login

Draw.io:

https://www.draw.io

Exercise 1

Speedy is a delivery company that wants to offer its top management a new service for creating customer reports.

The company already provides some services in this area, but the reports produced are not customizable.

For this new service, a data analyst is available to the managers to understand their needs and develop the new report.

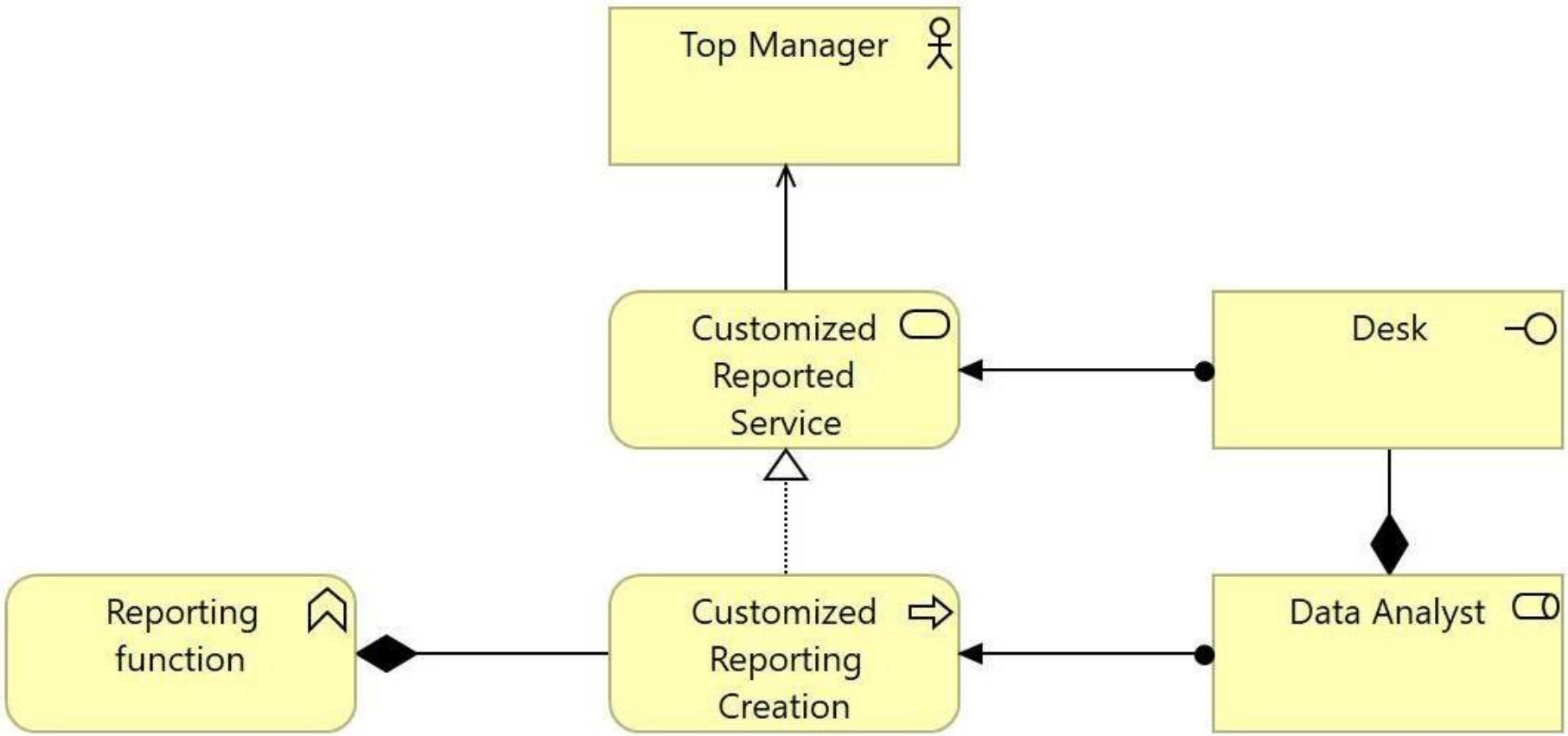
Exercise 1 – Key Elements

Speedy is a delivery company that wants to offer its top management a new service for creating customer reports.

The company already provides some services in this area, but the reports produced are not customizable.

For this new service, a data analyst is available to the managers to understand their needs and develop the new report.

Solution


Top Manager
Customized Reported Service
Desk
Reporting function
Customized Reporting Creation
Data Analyst

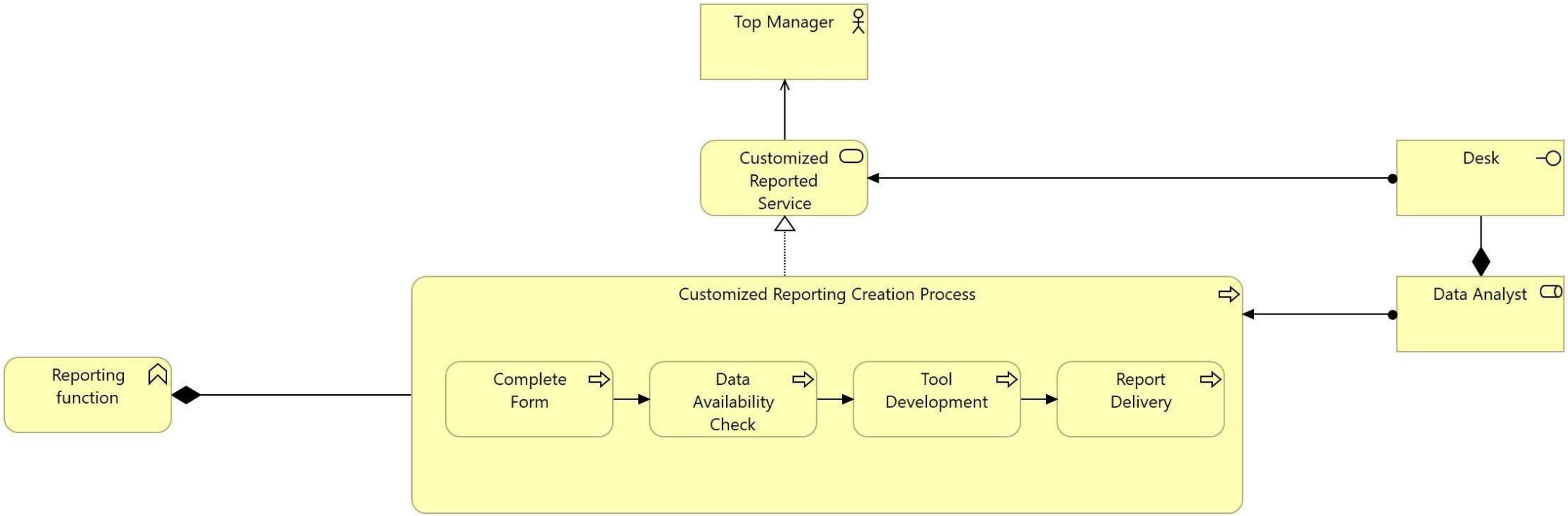
Exercise 1

To achieve this, the data analyst will require you to fill out a form, verify the available data, develop the tools needed to produce the report, and finally deliver that report to the manager.

Step:

- Filling out a form
- check the available data
- Develop the tools needed to produce the report
- deliver this report to the manager.

Solution


Top Manager
Customized Reported Service
Desk
Customized Reporting Creation Process
Data Analyst
Reporting function
Complete Form
Data Availability Check
Tool Development
Report Delivery

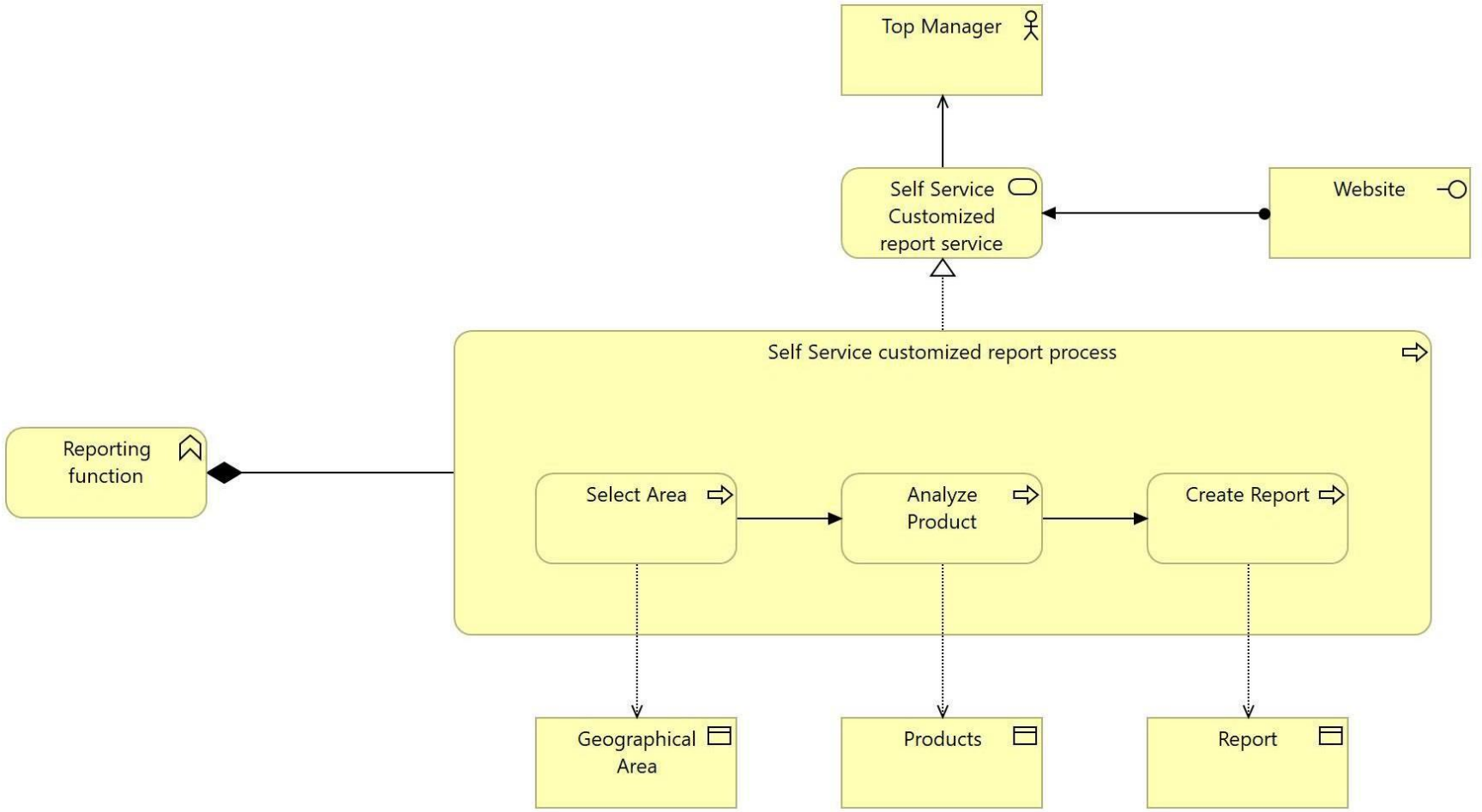
Exercise 1 – additional version (Self Service)

Speedy is a delivery company that wants to offer its top management a new service for creating customer reports.

The company already offers some services in this direction, but the reports produced are not customizable.

The new service is offered through a **web interface** that will allow the manager to first **select a geographical area**, then to **analyze the different products** and finally to **create a report**.

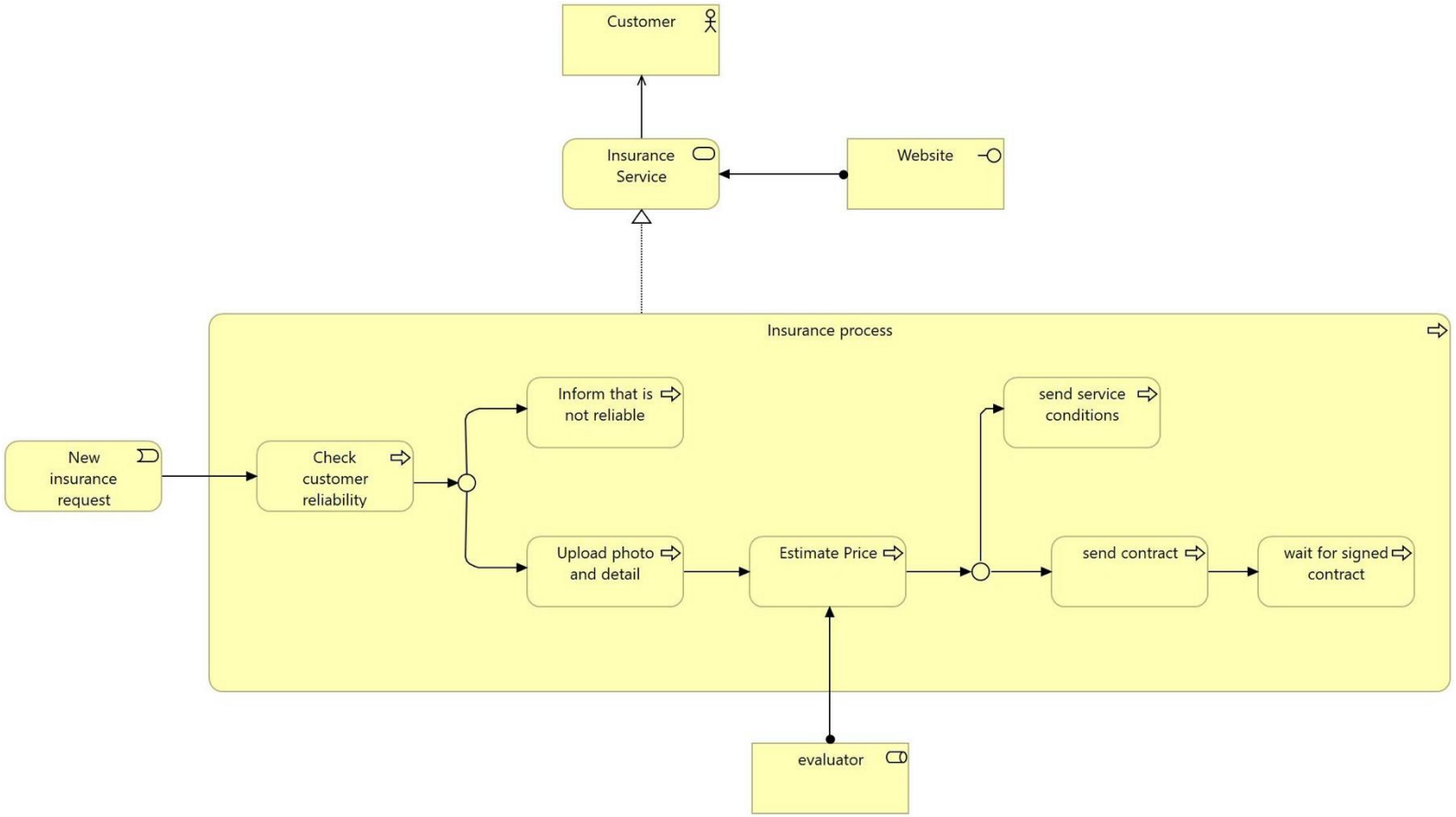
Soluzione – Self Service



Exercise 2

C is an insurance company that wants to offer a new insurance service for small items (under $2,000), managed entirely online for trustworthy customers.
The new process begins when a new insurance request is received. Then, the customer's credentials and past history are verified.
If the customer is considered trustworthy, they can upload a photo of the item along with the relevant details (serial number, purchase date).
At this point, an employee estimates the value of the item and, if it is below $2,000, the system sends a contract to the customer and waits for the signed copy.
If the estimated value exceeds $2,000, the system sends a message to the customer explaining the service conditions.

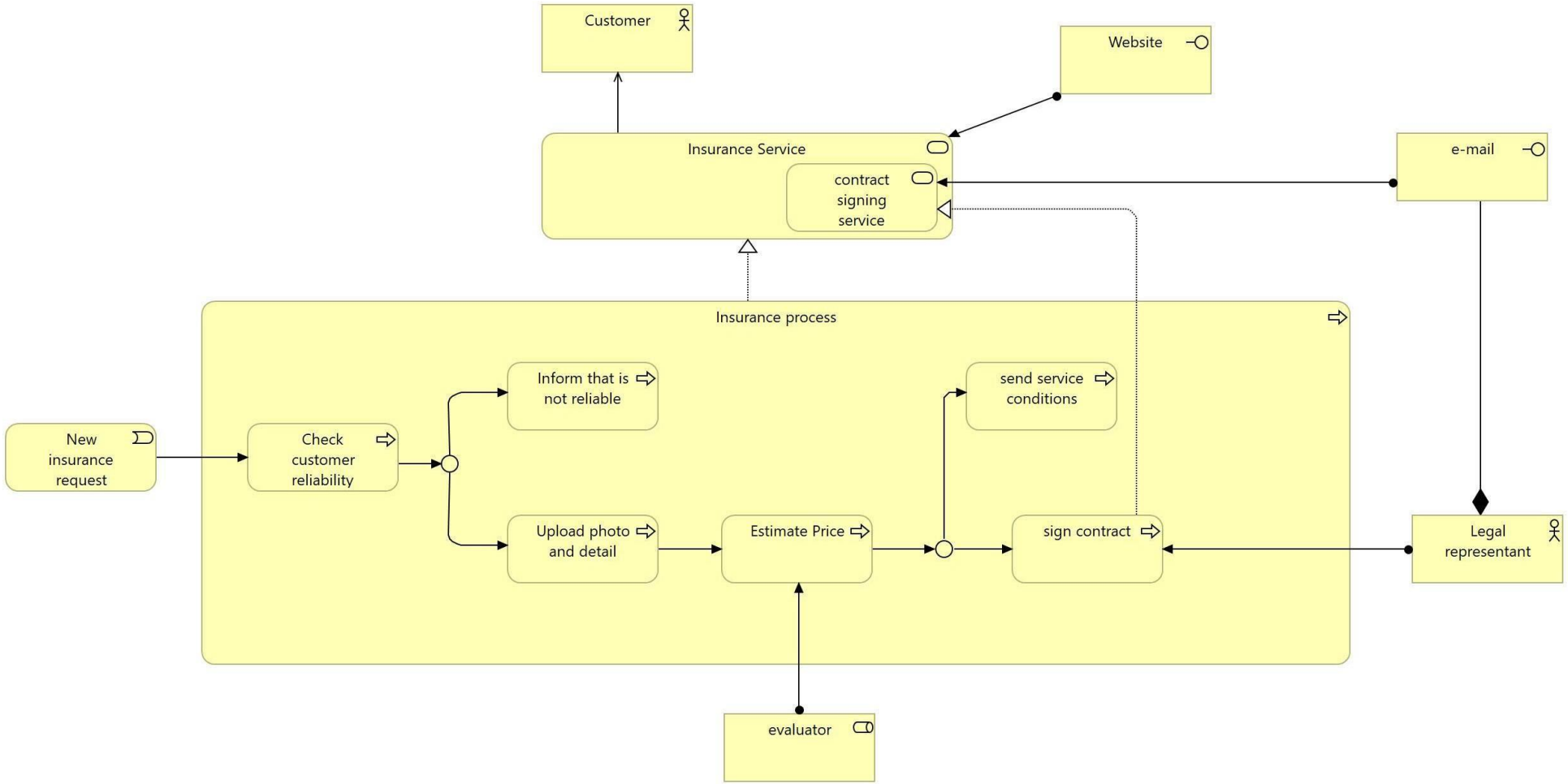
Solution


Customer
Insurance Service
Website
Insurance process
New insurance request
Check customer reliability
Inform that is not reliable
Upload photo and detail
Estimate Price
send service conditions
send contract
wait for signed contract
evaluator

Exercise 2 – alternative version

IC is an insurance company that wants to offer a new insurance service for small items (under $2,000), managed entirely online for trustworthy customers.
The new process begins when a new insurance request is received. Then, the customer's credentials and past history are verified.
If the customer is considered trustworthy, they can upload a photo of the item along with the relevant details (serial number, purchase date).
At this point, an employee estimates the value of the item and, if the item is worth less than $2,000, the customer and the legal representative begin exchanging documents via email to produce the signed contract.
If the estimated value exceeds $2,000, the system sends a message to the customer explaining the service conditions.

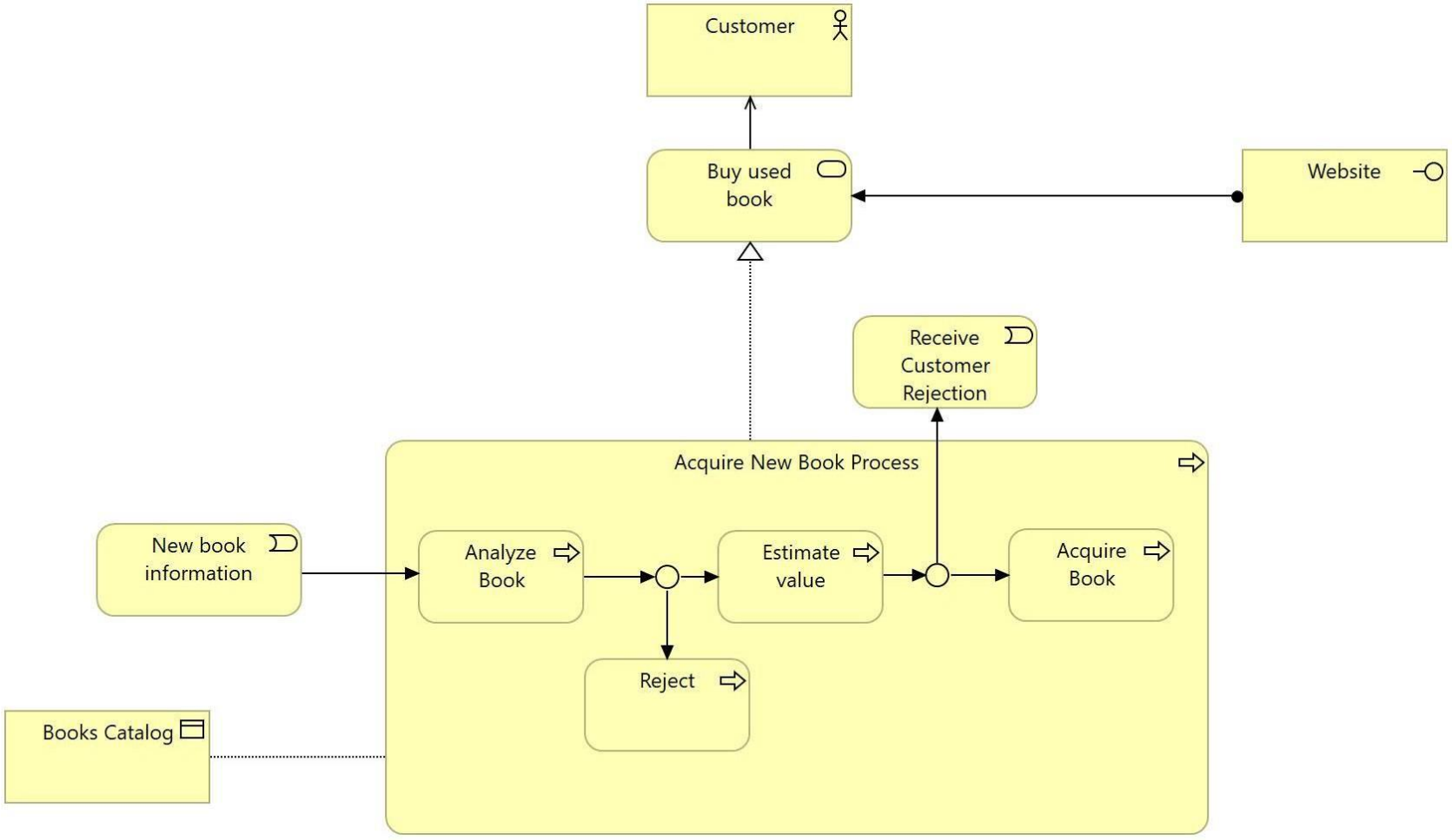
Solution


Customer
Website
Insurance Service
contract signing service
e-mail
Insurance process
Inform that is not reliable
send service conditions
New insurance request
Check customer reliability
Upload photo and detail
Estimate Price
sign contract
Legal representant
evaluator

Ubooks

UBooks buys and resells used books. Every time it purchases a book from a customer, it must ensure that it can resell it at a higher price to avoid losses. The process through which it buys a book is as follows: first, the user logs in, then enters the book information and details about its condition. UBooks analyzes the book to determine its condition.
If the condition is not deemed sufficient, the book is rejected. Otherwise, it estimates the value and presents an offer to the customer. If the customer accepts the offer, the book is acquired.

Solution

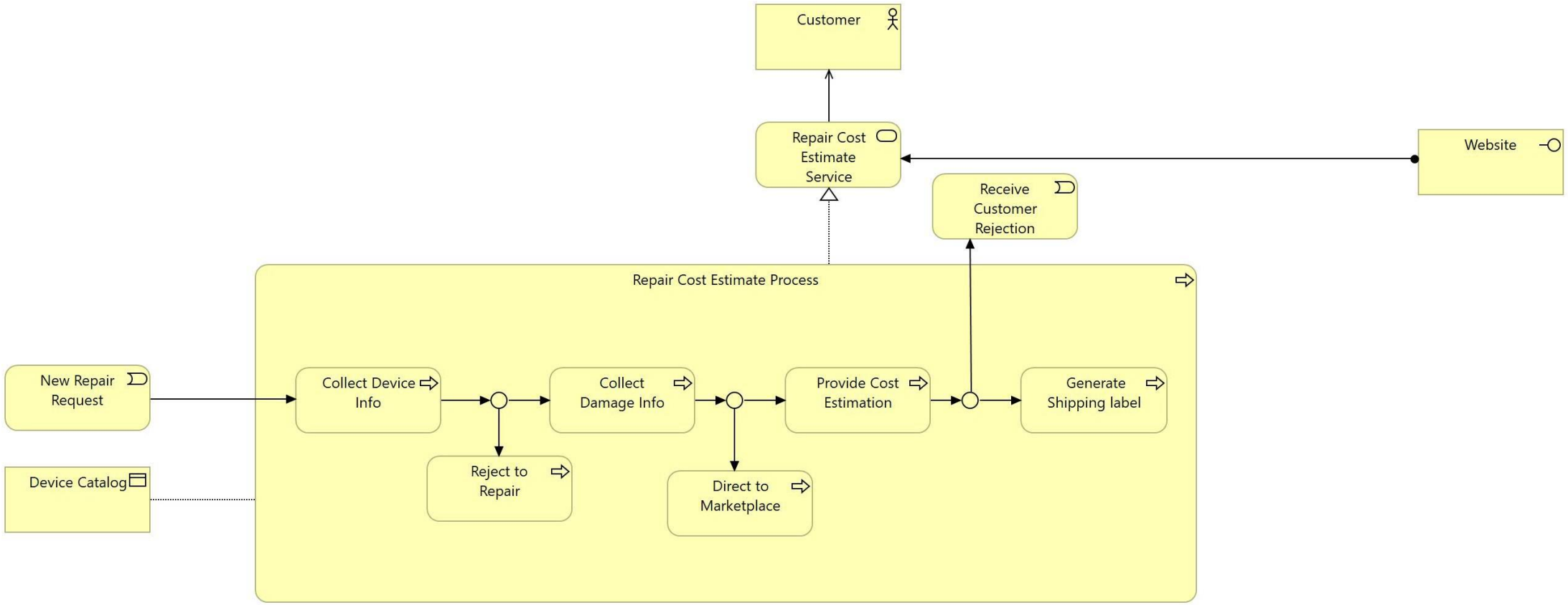

Customer
Buy used book
Website
Receive Customer Rejection
Acquire New Book Process
New book information
Analyze Book
Estimate value
Acquire Book
Reject
Books Catalog

I-REP

I-REP is a new company that offers repair services for smartphones, laptops, and other small electronic devices. A key feature of its value proposition is the ability for customers to enter the details of the device to be repaired and the type of damage through a mobile app, and immediately receive a cost estimate before sending in the device.

The company has implemented a new process that begins by collecting information about the device. If the device is on the list of those supported by the company, information about the damage is then gathered. At this point, there are two options: if the repair and processing costs exceed the value of the device, the user is informed and directed to a partner marketplace to purchase a similar product. Otherwise, a cost estimate range is provided, and the user has two options: reject it, or accept it, pay a deposit, enter their address details, and print the prepaid shipping label.

Solution

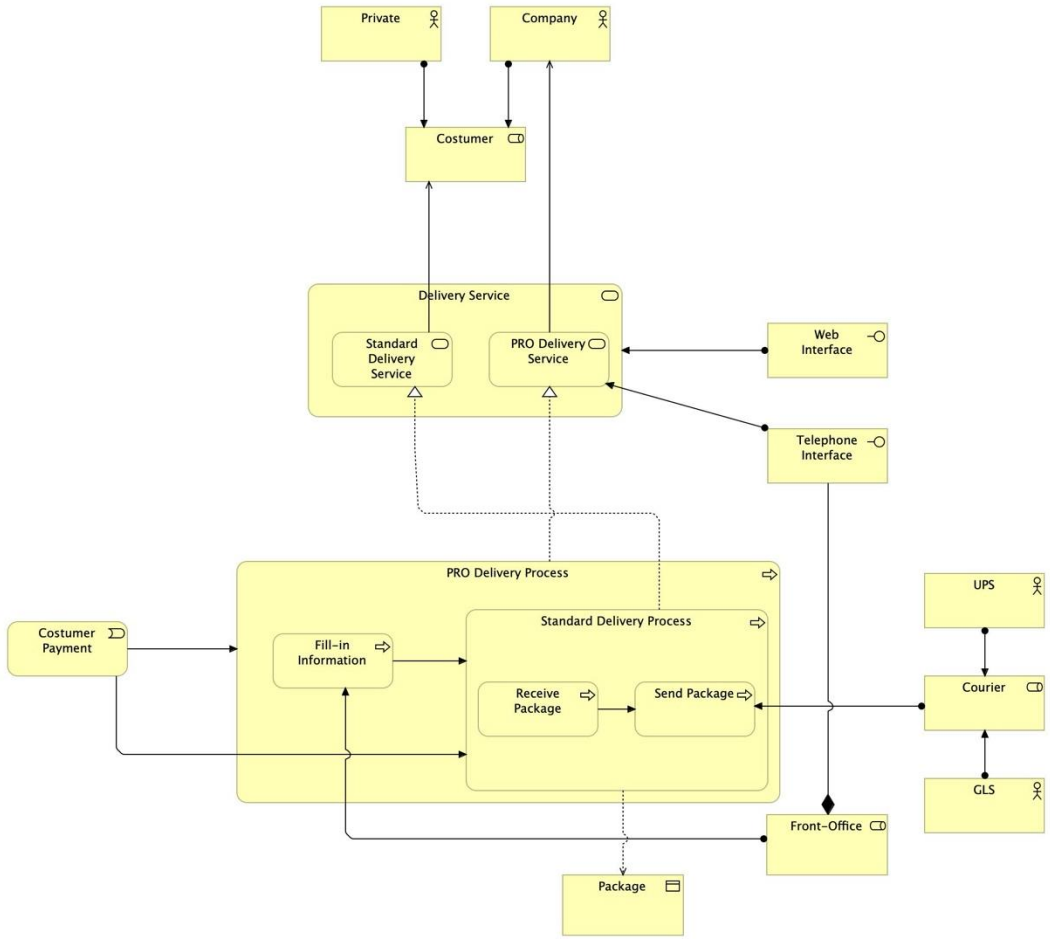

Customer
Repair Cost Estimate Service
Website
Receive Customer Rejection
Repair Cost Estimate Process
New Repair Request
Collect Device Info
Collect Damage Info
Provide Cost Estimation
Generate Shipping label
Device Catalog
Reject to Repair
Direct to Marketplace

Exercise - ConsegnaMi

The ConsegnaMi company operates in the Milan area and offers **various delivery services** for **individuals** and **companies**. The company offers a **standard delivery service** to all its customers which is offered via the **company's website**. The service consists of two steps: **receiving the package** from the customer and **shipping it**. The shipment is made by a **courier**, which can be **GLS** or **UPS**. The service is made available when **payment is received from the customer**.

The same company also offers a **PRO shipping service**, only to **companies**, always offered via a **web interface**. The PRO shipping service consists of the same steps as the standard service with the addition of an a priori step, where the **front office** helps, by **telephone**, the client company in **filling in the necessary data**. Again, the process is **triggered after the customer has paid**.

Solution


Private
Company
Costumer
Delivery Service
Standard Delivery Service
PRO Delivery Service
Web Interface
Telephone Interface
PRO Delivery Process
Standard Delivery Process
Costumer Payment
Fill-in Information
Receive Package
Send Package
UPS
Courier
GLS
Front-Office
Package

Exercise – ViaggiaMi

ViaggiaMi is a travel agency in Milan that offers, among other travel planning functions, a business travel organization service for employees of the Politecnico di Milano. The service is followed by a ViaggiaMi employee who interacts via email with the employees of Politecnico. The service consists of two steps, first the creation of a travel proposal, then the definition of the trip itself. The service is made available after the business trip has been authorized.

In addition, ViaggiaMi also offers an event tracking service, among the travel planning functions. In this case, the ViaggiaMi employee notifies travelers by phone in the event of significant events during the trip. The service is offered only after the traveler has left and consists of a first step, where events are monitored, and, possibly, a second step where travelers are notified.

Esercizio - ViaggiaMi

